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(Major Project)

Automated and Scalable Platform for Subscription Management with Alerts

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DECLARATION

This is to certify that the material embodied in this Major Project Dissertation titled **"Automated and Scalable Platform for Subscription Management with Alerts"**, submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**, is based on my original work.

It is further certified that this Major Project Dissertation has not been submitted, in full or in part, to this or any other university for the award of any other degree or diploma. All sources of information and assistance received during the course of this work have been duly acknowledged at the appropriate places.

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ABSTRACT

Subscription-based services have become a significant part of modern digital lifestyles. Individuals frequently subscribe to multiple services such as streaming platforms, cloud storage, software tools, utilities, and digital memberships. Managing these subscriptions manually often leads to missed renewal dates, unexpected expenses, and a lack of visibility into recurring financial commitments.

The project **"Automated and Scalable Platform for Subscription Management with Alerts (Sub Track AI)"** proposes a centralized system that enables users to efficiently track, manage, and analyse their subscriptions. The platform is designed using a **microservices-based architecture** to ensure scalability, modularity, and maintainability.

The system allows users to add and manage subscriptions, monitor billing cycles, calculate monthly recurring costs, and receive automated alerts before renewal dates. The platform integrates modern backend technologies including **Java Spring Boot microservices**, **JWT-based authentication**, and **REST APIs** for secure communication between services.

The mobile interface is developed using **React Native**, providing users with a convenient way to manage subscriptions, view analytics dashboards, and receive notifications through a mobile application. The backend services communicate through an API Gateway and are containerized using **Docker** to simulate enterprise-level deployment environments.

The expected outcome of the project is to enhance users' financial awareness, minimize unnecessary recurring expenses, and demonstrate the practical implementation of **microservices architecture, event-driven systems, and automation** for solving real-world financial management problems.

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CHAPTER 1: Introduction

1.1 Project Overview

In the modern digital ecosystem, subscription-based services have become an essential part of daily life. Individuals frequently subscribe to various platforms such as video streaming services, cloud storage solutions, productivity tools, software applications, online learning platforms, and digital memberships. These services provide convenience and flexibility by offering continuous access to digital content and tools through recurring payments.

However, as the number of subscriptions increases, managing them manually becomes difficult for users. Many users forget renewal dates, lose track of active subscriptions, or continue paying for services that they no longer use. This often leads to unnecessary expenses and a lack of transparency regarding recurring financial commitments. Most existing expense management applications focus mainly on tracking one-time transactions and do not provide dedicated features for managing recurring subscriptions effectively.

To address this issue, the project **"Automated and Scalable Platform for Subscription Management with Alerts (Sub Track AI)"** proposes a centralized system that helps users efficiently track and manage their subscriptions. The platform allows users to store details of their subscriptions, monitor billing cycles, calculate monthly recurring costs, and receive timely alerts before renewal dates.

The system is designed using a **microservices-based architecture**, which divides the application into independent services responsible for specific functionalities such as authentication, user management, subscription tracking, and notifications. This architecture improves scalability, flexibility, and maintainability while enabling the system to simulate real-world enterprise applications.

The backend services are developed using **Java Spring Boot**, which provides robust frameworks for building scalable REST-based microservices. A mobile application built using **React Native** serves as the user interface, allowing users to manage subscriptions and view analytics through a convenient and responsive mobile platform.

Overall, the Sub Track AI platform aims to improve financial awareness, simplify subscription management, and demonstrate the practical implementation of modern software engineering practices such as microservices architecture, containerization, and distributed system design.

1.2 Objectives of the Project

The primary objectives of this project are to design and develop an intelligent platform that enables users to efficiently manage their subscription-based services. The main objectives of the project are as follows:

1. **To develop a centralized subscription management system** that allows users to store and monitor all their subscriptions in one place.

2. **To provide automated alerts and reminders** before subscription renewal dates in order to help users avoid unexpected charges.

3. **To track and analyse recurring expenses** by calculating monthly or yearly subscription costs and presenting them through simple analytics.

4. **To implement a scalable microservices architecture** that separates system functionalities into independent services for better maintainability and scalability.

5. **To design a user-friendly mobile application** using React Native that allows users to easily manage subscriptions and receive notifications.

6. **To demonstrate the application of modern software development technologies** such as Spring Boot, REST APIs, JWT authentication, Docker containerization, and distributed system design.

7. **To improve users' financial awareness and control over recurring payments** by providing visibility into their subscription spending patterns.

Chapter 2: Problem Statement

2.1 Problem Definition

In recent years, subscription-based business models have become increasingly popular across various digital platforms. Services such as video streaming, music streaming, cloud storage, software tools, online learning platforms, and digital memberships operate primarily on recurring subscription payments. These services provide users with convenient access to content and applications without requiring large upfront costs.

However, as users subscribe to multiple services over time, managing these subscriptions becomes challenging. Many individuals often lose track of their active subscriptions and renewal dates. In several cases, users continue paying for services they no longer actively use because the payments are automatically deducted on a monthly or yearly basis. This results in unnecessary financial expenditure and poor visibility of recurring financial commitments.

Another significant challenge is the absence of a centralized platform where users can view and manage all their subscriptions in one place. Existing expense management systems mainly focus on tracking daily transactions such as purchases and bills, but they generally lack dedicated mechanisms for monitoring recurring subscription payments. As a result, users must manually track their subscriptions through emails, bank statements, or separate reminders, which is inefficient and prone to errors.

Additionally, most traditional systems do not provide intelligent notification mechanisms that alert users before a subscription renewal occurs. Without timely reminders, users may miss the opportunity to cancel or modify a subscription before the next billing cycle begins.

Therefore, there is a clear need for a smart and scalable system that can efficiently manage subscription-based services. Such a system should allow users to track their subscriptions, analyse recurring expenses, and receive automated alerts before renewal dates. The **Sub Track AI platform** aims to solve this problem by providing a centralized and automated solution that simplifies subscription management and enhances financial awareness among users.

2.2 Scope of the Work

The scope of this project involves designing and developing a subscription management platform that helps users efficiently organize and monitor their recurring digital subscriptions. The system focuses on providing essential functionalities that allow users to track their subscriptions, receive alerts, and analyse their spending patterns.

The platform will enable users to add and manage details of their subscription services, including the service name, billing cycle, renewal date, and subscription cost. Users will

be able to view all their subscriptions in a centralized dashboard, making it easier to monitor their recurring financial commitments.

Another important aspect of the system is the implementation of automated notifications. The system will generate alerts before subscription renewal dates so that users can decide whether to continue, cancel, or modify their subscriptions. These alerts may be delivered through in-app notifications or other communication channels.

The project also includes the development of a **microservices-based architecture** to improve system scalability, maintainability, and modularity. Different services such as authentication, user management, subscription management, and notification services will operate independently while communicating through APIs.

The mobile interface developed using **React Native** will provide users with a convenient platform to interact with the system. Through the mobile application, users will be able to manage their subscriptions, view analytics, and receive alerts in real time.

However, the current scope of the project does not include direct integration with banking systems, payment gateways, or automatic financial transactions. Instead, the system focuses primarily on subscription tracking, alert generation, and expense analysis. Future versions of the platform may extend these capabilities by integrating external financial services and implementing advanced data analytics features.

Overall, the scope of the project is to design a scalable and efficient subscription management platform that demonstrates the practical implementation of modern software engineering concepts while providing users with better control over their recurring digital subscriptions.

Chapter 3: Technology & System Details

3.1 Technology Stack

The development of the **Sub Track AI platform** involves the use of modern technologies that support scalable system design, efficient data management, and mobile accessibility. The system follows a **microservices-based architecture**, where different services handle specific functionalities independently. The major technologies used in the project are described below.

Layer	Technology
Backend	Java Spring Boot
API Gateway	Spring Cloud Gateway
Mobile Application	React Native
Database	MySQL
DevOps	Docker

Table 3.1: Layers and Technology

Backend (Java Spring Boot): The backend services of the system are developed using **Java Spring Boot**, a widely used framework for building enterprise-level applications. Spring Boot provides built-in support for REST APIs, dependency injection, security integration, and database connectivity. It enables rapid development of microservices that are modular, scalable, and easy to maintain.

API Gateway (Spring Cloud Gateway): Spring Cloud Gateway acts as the **central entry point** for all client requests. It routes incoming requests from the mobile application to the appropriate backend microservice. The gateway also helps in handling cross-cutting concerns such as request routing, security filtering, and load balancing.

Mobile Application (React Native): The user interface of the system is developed using **React Native**, a cross-platform mobile application framework that allows developers to build mobile apps using JavaScript and TypeScript. React Native enables the creation of

responsive and user-friendly mobile interfaces while sharing code across Android and iOS platforms.

Database (MySQL): The system uses **MySQL** as the relational database management system. MySQL stores important data such as user accounts, subscription details, billing cycles, and notification records. It provides reliable data storage, efficient query processing, and strong support for relational data models.

DevOps (Docker): Docker is used to containerize the application services. Containerization ensures that each microservice runs in an isolated environment with its required dependencies. This simplifies deployment, improves portability, and ensures consistent behaviour across development and production environments.

3.2 Tools / Platforms Used

Several development tools and platforms are used to design, implement, and manage the project efficiently.

Spring Boot: Spring Boot is used for developing the backend microservices. It simplifies application development by providing pre-configured components and integration with Spring frameworks such as Spring Security and Spring Data JPA.

React Native: React Native is used for building the mobile application interface. It allows developers to create mobile applications using JavaScript and React components, making development faster and more efficient.

MySQL: MySQL is used for managing structured data within the system. It stores information related to users, subscriptions, billing details, and notification history.

Docker: Docker is used for containerizing the application components. Each service runs in a separate container, which ensures consistent deployment and simplifies system management.

GitHub: GitHub is used as a version control platform for managing the source code of the project. It allows developers to track code changes, collaborate efficiently, and maintain version history.

3.3 System Architecture

The Sub Track AI platform follows a **microservices architecture**, where the application is divided into multiple independent services. Each service is responsible for a specific domain of functionality and communicates with other services through REST APIs.

The architecture begins with the **mobile application**, which acts as the primary user interface. Users interact with the system through the mobile app to perform operations such as authentication, adding subscriptions, viewing analytics, and receiving notifications.

All client requests from the mobile application are first sent to the **API Gateway**. The API Gateway acts as a central routing mechanism that directs requests to the appropriate backend service. It also helps in handling authentication checks, request filtering, and load balancing.

The backend consists of multiple microservices, each responsible for a specific functionality:

1) **Authentication Service:** Handles user login, registration, and JWT-based authentication.
2) **User Service:** Manages user profiles, account information, and preferences.
3) **Subscription Service:** Handles subscription creation, updates, billing cycles, and subscription analytics.
4) **Notification Service:** Generates alerts and notifications related to subscription renewals and important updates.

All services interact with the **MySQL database**, which stores persistent data such as user records, subscription details, billing schedules, and notification logs.

The architecture flow of the system can be represented as follows:

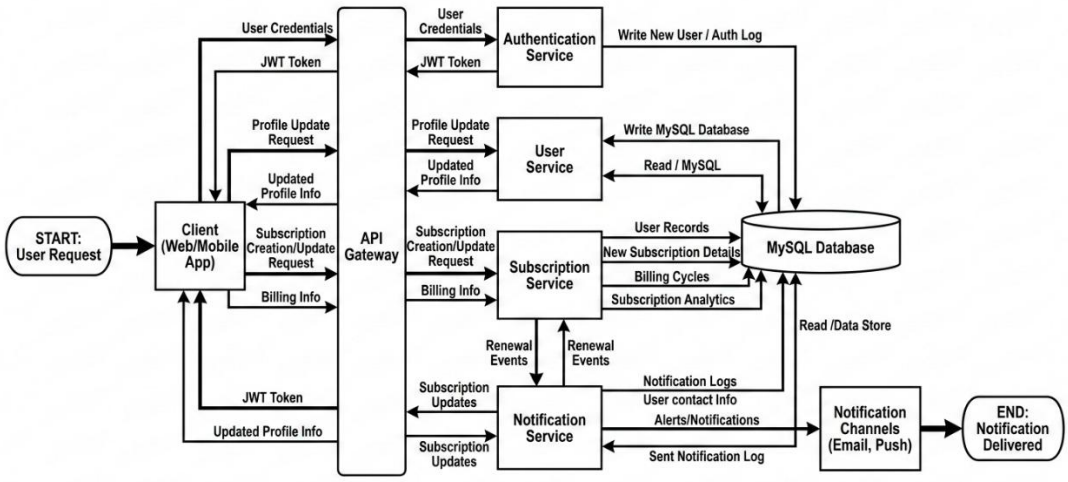

Fig. 3.1: Data Flow Diagram

This modular architecture improves scalability, maintainability, and system reliability. Since each service operates independently, new features can be added or modified without affecting the entire system. Additionally, the use of containerization through Docker allows each service to be deployed and managed efficiently in different environments.

Chapter 4: Progress Status

4.1 Work Completed till Date

The development of the **Sub Track AI platform** has progressed through several important stages, including system design, backend development, database setup, and the initial implementation of the mobile application. The work completed so far focuses on establishing the core architecture of the system and implementing the primary functionalities required for subscription management.

One of the initial steps in the project was the **design of the microservices architecture**. The system was structured into multiple independent services to ensure modularity, scalability, and easier maintenance. Each service was assigned a specific responsibility, such as authentication, user management, subscription handling, and notification processing.

The **API Gateway** was implemented to act as a centralized entry point for all incoming requests from the mobile application. The gateway manages request routing and directs each request to the appropriate backend service, ensuring communication within system.

The **authentication service** was developed to handle user registration, login, and secure access to the platform. This service implements **JWT (JSON Web Token) based authentication**, which ensures that only authorized users can access protected resources.

Another major component completed during this stage is the **subscription management service**, which allows users to perform basic CRUD (Create, Read, Update, Delete) operations on subscription data. This enables users to add new subscriptions, modify existing subscription details, and remove subscriptions that are no longer required.

The **MySQL database** has been configured to store application data, including user information, subscription details, and other related records.

In addition to backend development, the **initial version of the mobile application** has been developed using React Native. The mobile interface provides users with basic screens for login, viewing subscription data, and interacting with the system.

Overall, the work completed so far establishes the core infrastructure of the system and prepares the platform for additional features such as automated notifications, analytics dashboards, and advanced subscription tracking capabilities.

4.2 Milestones Achieved

Several important milestones have been achieved during the development phase of the project. These milestones mark significant progress toward the completion of the subscription management platform.

One of the major milestones achieved is the successful implementation of the **authentication module**, which enables secure user login and access control through JWT-based authentication. This module forms the foundation of system security and ensures that only registered users can access the application services.

Another important milestone is the development of the **subscription management service**, which allows users to add, update, view, and delete subscription information. This functionality forms the core feature of the platform and enables users to organize their subscription details in a centralized system.

The development of the **mobile application prototype** represents another key milestone. The prototype demonstrates the basic interaction between the mobile interface and backend services, allowing users to log in and access initial application features.

Additionally, the system has been **containerized using Docker**, which allows all services to run within isolated containers.

These milestones indicate that the project has successfully completed the foundational stages of development and is now ready to move toward advanced features and system enhancements.

4.3 Screenshots / Prototype

This section presents the initial prototype of the **Sub Track AI mobile application**. The screenshots demonstrate the user interface and provide a visual representation of the system functionality developed so far.

The **Login Screen and Signup Screen** allows users to securely authenticate themselves by entering their credentials.

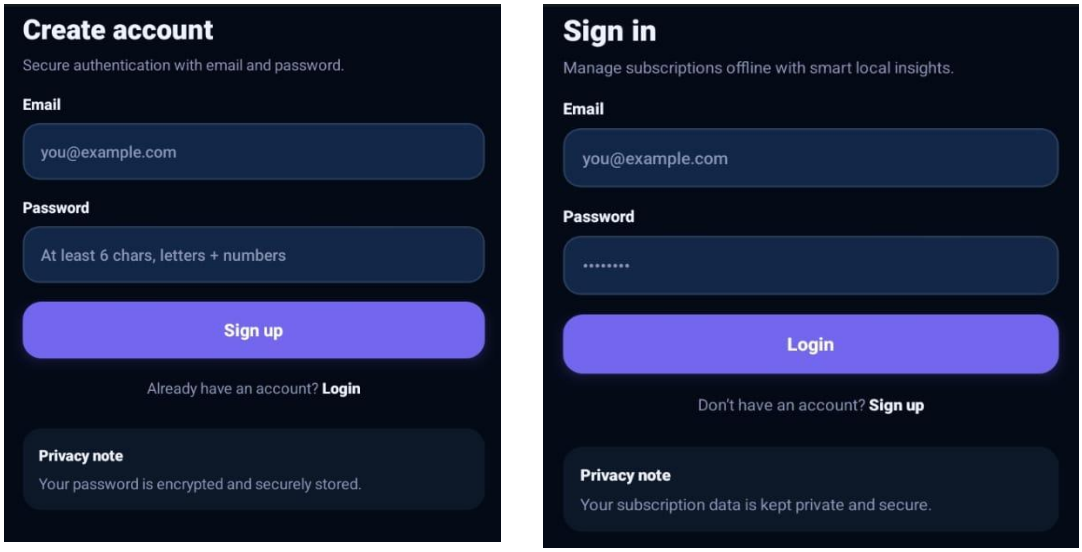

Fig 4.1: Signup and Login Screen

The **Dashboard Screen** provides an overview of the user's subscription data. It displays important information such as active subscriptions, upcoming renewals, and summary statistics related to recurring expenses.

Fig 4.2: Dashboard Screen Shot

The **Subscription Management Screen** allows users to add new subscriptions and view existing ones. Users can enter details such as the subscription name, cost, billing cycle, and renewal date.

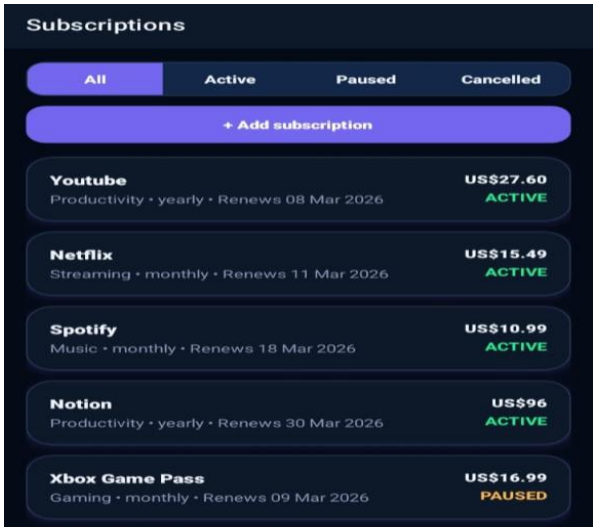

Fig 4.3: Subscription Management Page Screen Shot

Screenshots of these interfaces will be included in this section to demonstrate the current progress of the mobile application and provide visual documentation of the system prototype.

Chapter 5: Challenges and Solutions

5.1 Technical Challenges

During the development of the **Sub Track AI platform**, several technical challenges were encountered while designing and implementing the system. Since the project follows a **microservices-based architecture**, careful planning and technical implementation were required to ensure scalability, security, and efficient communication between services.

One of the primary challenges was **designing a scalable microservices architecture**. Unlike traditional monolithic systems, microservices divide the application into smaller independent services that must communicate with each other. Ensuring that each service performs its specific task while maintaining system stability required proper service design, clear API definitions, and efficient request routing mechanisms.

Another challenge involved **implementing secure authentication and authorization mechanisms**. Since the system allows users to access sensitive personal information related to their subscriptions, it was important to implement a secure authentication system. Managing user login sessions, protecting APIs, and ensuring that only authorized users can access system resources required the integration of secure authentication protocols.

Additionally, **managing communication between multiple services** presented another technical challenge. In a microservices architecture, services need to interact with each other frequently. Without proper routing and communication management, this could lead to complex service dependencies and inefficient request handling.

These challenges required the adoption of suitable technologies and system design strategies to ensure that the platform remains secure, scalable, and efficient.

5.2 Solutions Implemented

To overcome the technical challenges encountered during development, several effective solutions and technologies were implemented in the system architecture.

To manage communication between services and simplify request routing, **Spring Cloud Gateway** was used as the API Gateway. The API Gateway acts as a centralized entry point for all incoming client requests. It routes these requests to the appropriate microservice, reducing the complexity of direct service-to-service communication and improving overall system organization.

For secure user authentication, the system implements **JWT (JSON Web Token) based security**. JWT tokens are generated when users successfully log in to the system and are used to verify the identity of users when they access protected APIs. This method ensures secure communication between the client and server while preventing unauthorized access to application resources.

To simplify deployment and ensure consistent system behaviour across different environments, the application services were **containerized using Docker**. Docker containers package each service with its required dependencies, allowing the application to run reliably across development, testing, and deployment environments. Containerization also improves system scalability by allowing services to be deployed and managed independently.

By implementing these solutions, the system achieves a stable and scalable architecture that supports secure authentication, efficient service communication, and flexible deployment. These technologies help ensure that the **Sub Track AI platform** can operate effectively while maintaining high levels of performance and reliability.

Chapter 6: Future Plan & Expected Outcomes

6.1 Upcoming Work Plan

Although the foundational components of the **Sub Track AI platform** have been developed, several important features and improvements are planned for the next phase of the project. The upcoming work will focus on enhancing system automation, improving user experience, and implementing advanced analytical capabilities.

One of the major planned enhancements is the implementation of an **asynchronous notification processor**. This component will monitor subscription events such as upcoming renewal dates and automatically trigger notifications for users. By using asynchronous processing mechanisms, the system will be able to handle notifications efficiently without affecting the performance of the core application services.

Another important improvement will involve the integration of **advanced analytics features**. The analytics module will provide users with insights into their subscription spending patterns, including monthly expenses, category-based analysis, and trends in recurring payments. These insights will help users better understand their financial commitments and make informed decisions regarding their subscriptions.

The project also plans to integrate **push notification services** within the mobile application. Push notifications will allow users to receive real-time alerts about upcoming subscription renewals, payment reminders, and important updates directly on their mobile devices. This feature will significantly enhance user engagement and ensure that users do not miss important subscription events.

In addition to these features, further improvements will be made to the user interface, system performance, and overall application stability. The development team will also focus on refining the existing modules and ensuring seamless communication between all microservices.

6.2 Expected Deliverables

At the completion of the project, the **Sub Track AI platform** is expected to deliver a fully functional and scalable subscription management system that helps users efficiently track and manage their recurring digital subscriptions.

The final deliverable will include a **fully functional subscription management platform** built using a microservices-based architecture. The system will provide users with the ability to add, update, and monitor subscription services through a centralized platform.

Another key deliverable will be the **mobile application**, developed using React Native, which will allow users to conveniently access the system from their smartphones. The

mobile application will provide features such as subscription tracking, expense summaries, and notification alerts through an intuitive and user-friendly interface.

Additionally, the platform will include an **automated alert and notification system** that reminds users about upcoming subscription renewals. This feature will help users avoid unnecessary charges and improve their financial awareness by providing timely reminders.

Overall, the completed project will demonstrate the practical implementation of **modern software engineering practices**, including microservices architecture, containerized deployment, and automated notification systems. The system aims to provide users with better control over their subscription expenses while showcasing a scalable and efficient solution to the challenges of managing recurring digital services.

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